SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 60%

QUESTIONS: 6 **Total Marks:60**

Annexure A

Application Based Questions

Code of Conduct:

I Deepa lakshana.B (192471054)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

Actors

- Guest
- Manager
- Payment Service
- Administrator (Missing Actor)

Use Cases

Guest

- Register/Login
- Search Rooms
- View Room Details
- Make Reservation
- Cancel Booking
- Make Payment
- View Booking Status

Manager

- Allocate Rooms
- Manage Reservations
- Generate Reports

Payment Service

- Process Payment
- Verify Payment

Administrator

- Manage Users
- Update Room Availability
- Manage Hotel Information

Relationships

- **<<include>>**
    - Make Reservation → Search Rooms
    - Make Reservation → Process Payment
    - Cancel Booking → Verify Booking
- **<<extend>>**
    - Cancel Booking extends Reservation
- **Generalization**
    - User
        - Guest
        - Manager
        - Administrator

Use Case Diagram (Text Form)

```
              User
          /     |     \
     Guest    Manager   Administrator
```

Guest:

Login
Search Rooms
View Room
Reserve Room
Cancel Booking
Make Payment
Track Booking

Manager:
Allocate Room
Manage Reservation
Generate Reports

Administrator:
Manage Users
Update Rooms

Payment Service:
Process Payment

Reserve Room
<<include>>
Search Room
Process Payment

Cancel Booking
<<extend>>
Reserve Room

2. Class Diagram Design

Classes
Student
Attributes

- studentId
- name
- department

Methods

- registerCourse()
- viewResult()

Faculty
Attributes

- facultyId
- name
- designation

Methods

- teachCourse()
- evaluateExam()

Course
Attributes

- courseId
- courseName
- credits

Methods

- assignFaculty()

Department
Attributes

- deptId
- deptName

Methods

- addCourse()

Examination
Attributes

- examId
- examDate
- marks

Methods

- conductExam()
- publishResult()

Relationships

- Department **1..*** Course (Aggregation)
- Student **many-to-many** Course (Association)
- Faculty **1..*** Course (Association)
- Course **1..*** Examination (Composition)
- Person → Student, Faculty (Inheritance)

3. Sequence Diagram Evaluation

Objects

- Customer
- Vehicle Search
- Booking System
- Payment Gateway
- Database

Sequence

Customer → Search System : Search Vehicle

Search System → Database : Check Availability

Database → Search System : Available Vehicles

Search System → Customer : Display Vehicles

Customer → Booking System : Select Vehicle

Booking System → Payment Gateway : Payment Request

Payment Gateway → Customer : Enter Payment Details

Customer → Payment Gateway : Confirm Payment

Payment Gateway → Booking System : Payment Success

Booking System → Database : Save Booking

Database → Booking System : Booking Saved

Booking System → Customer : Rental Confirmation

4. State Diagram Construction

States

- Red Signal
- Green Signal

- Yellow Signal
- Emergency Mode
- Maintenance Mode

State Diagram (Text Form)
Start

↓

Red Signal

↓

Timer Expires

↓

Green Signal

↓

Timer Expires

↓

Yellow Signal

↓

Timer Expires

↓

Red Signal

Emergency Vehicle

↓

Emergency Mode

↓

Resume

↓

Red Signal

Maintenance Request

↓

Maintenance Mode

↓

Maintenance Complete

↓

Red Signal

Events

- Timer Expiry
- Emergency Vehicle Detection
- Maintenance Request
- Resume Service

5. Component Diagram Design

Components

- Patient Management
- Appointment Scheduling
- Medical Records
- Billing
- Notification Service
- Database

Component Relationships

Patient Management
|
|
Appointment Scheduling
|
Medical Records
|
Billing
|
Notification Service
|
Database

Dependencies

- Appointment depends on Patient Management.
- Medical Records depend on Appointment.
- Billing depends on Medical Records.
- Notification Service sends SMS/Email after Billing.
- All components access the Database.

Interfaces

- IPatientService
- IAppointmentService

- IBillingService
- INotificationService

6. Deployment Diagram Design

Hardware Nodes

- IoT Waste Bin Sensor
- Edge Device
- Cloud Server
- Monitoring Dashboard
- Mobile Device

Software Artifacts

- Sensor Firmware
- Edge Processing Software
- Cloud Analytics
- Waste Monitoring Application
- Database

Deployment Diagram (Text Form)

```
+-------------------------+
| Mobile Device           |
| Waste Monitoring App    |
+-------------------------+
          |
       Internet
          |
+-------------------------+
| Cloud Server            |
| Analytics               |
| Database                |
+-------------------------+
```

```
           |
+-----------------------+
| Edge Device           |
| Edge Processing       |
+-----------------------+
           |
+-----------------------+
| IoT Waste Bin Sensors |
| Sensor Firmware       |
+-----------------------+
```

Communication Links

- IoT Sensors ↔ Edge Device
- Edge Device ↔ Cloud Server
- Cloud Server ↔ Monitoring Dashboard
- Mobile App ↔ Cloud Server

Advantages

- Real-time waste monitoring
- Faster data processing
- Reduced network traffic
- Scalable cloud architecture
- Remote monitoring and control

RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand